

NewsAnalyzer: Real-Time Grounded Fake News Detection on Traditional Chinese Social Media via Two-Stage Fact-Check Retrieval, Multi-Signal Metadata Fusion, and Consistency-Gated Local Semantic Analysis

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Abstract. Fake news spreads fast on Traditional Chinese social media like Facebook, LINE chats, and Threads. Real news and fake posts look the same in the feed, so readers cannot tell them apart. Large language models are poor judges for this job: they do not know new hoaxes, they believe smooth but fake writing, and they give scores with no proof behind them. Cloud APIs are also slow and send user data away. Therefore, this study builds **NewsAnalyzer**, a checker that ties every score to proof it actually found. Its retrieval is deliberately *two-stage*: a hard similarity band admits a claim as a verified verdict, while an adjacent *soft band* is escalated to the auxiliary LLM as a relevance question rather than silently returned as `not_found`; sibling claims in the same rumor family are cross-confirmed across the top- k candidates, and entity gating rejects matches that share only a generic term. Scoring combines website trust, feeling words, and post age, and enforces a hard cap at 25 points ($S \leq 25$) whenever counter-evidence against the claim is found. The auxiliary local LLM never makes the call: it answers relevance and summary questions under grammar-constrained JSON decoding, and its verdicts are accepted only when repeated sampling agrees. We test all 2,292 claims with 5-fold cross-validation under rewrites from light to heavy, including cluster-level splits that exclude near-duplicates of every test claim. The checker was right 95.3% of the time ($F_1 = 0.974$) on solved claims at record level and retains 90.8% under the stricter cluster-level splits; retrieval resolves in 1.48ms. A 48-article sweep of real Taiwanese mainstream reporting exposes and then eliminates three classes of false-positive recall that no offline benchmark reveals. A probe of live Cofacts activity over the trailing 30 days finds that no recent item classified as a rumor was a news-shaped hoax—all were spam, scam, or link-only posts, which the system rejects at 25 points. Against majority, TF-IDF, NewsGuard-style, and equal-capacity zero-shot LLM baselines

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(99.0% vs. 88.2% on a paired 500-claim probe, $\Delta\text{Acc} = +10.85$ pp, McNemar $p = 1.2 \times 10^{-14}$), it was more accurate while keeping article content and analysis on-device and issuing only bounded lookup queries (post title and at most a 300-character claim prefix) to public fact-check and search services.

Keywords: Fake news detection, Traditional Chinese social media, fact-checking census, metadata scoring, semantic retrieval, dynamic safety anchoring, edge inference.

1 Introduction

In Taiwan, digital news consumption is heavily concentrated within high-velocity social platforms such as Facebook, LINE group chats, and Threads. In these environments, legitimate news reporting, hyperpartisan commentary, deceptive content-farm clickbait, and unverified public health rumors circulate in identical visual formats, often separated by only a single scroll. Combating this infodemic directly at the point of consumption requires credibility assessments that are accurate, instantly accessible, and strictly protective of user browsing privacy.

Recent academic and commercial initiatives have increasingly turned to standalone Large Language Models (LLMs) as zero-shot truth arbiters. However, relying on brute-force LLM generation as the primary judge introduces critical structural failure modes:

1. **Temporal Blindness and Information Lag:** Pre-trained LLM weights reflect a historical knowledge cutoff and cannot reliably evaluate breaking rumors or localized viral hoaxes that emerged hours or days prior.
2. **Susceptibility to Stylistic Deception:** As generative AI tools lower the cost of producing polished propaganda, modern misinformation frequently exhibits formal syntax and authoritative phrasing. Unconstrained LLMs evaluate such text as plausible and credible due to its fluent surface structure.
3. **Absence of Auditable Provenance:** Brute-force LLMs output subjective score predictions without binding them to verified investigation archives, risking plausible-sounding hallucinations that undermine user trust.
4. **Latency and Privacy Penalties:** Dispatching live browser DOM contents and browsing history to third-party cloud LLM APIs incurs severe round-trip latencies (5–10 s) and creates significant privacy vulnerabilities.

To overcome these structural limitations, we introduce **NewsAnalyzer**, a grounded multi-signal verification framework tailored to Traditional Chinese social media. Rather than treating an LLM as an ungrounded oracle, NewsAnalyzer anchors its detection pipeline on an offline macro census of verified fact-checking records ($N = 2,292$) sourced from the collaborative Cofacts repository and accredited by the International Fact-Checking Network (IFCN). This empirical base is fused with rich metadata heuristics (domain reputation, sentiment volatility, clickbait valence, and temporal decay) and enforced by an asymmetric safety clamping mechanism that overrules speculative model generation when

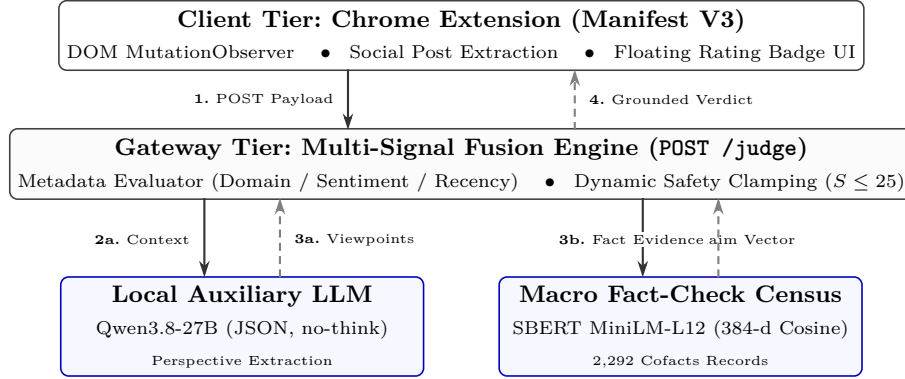


Fig. 1. Layered architecture of NewsAnalyzer: the client extension extracts live social DOM elements and communicates with the local gateway, which prioritizes empirical fact-check census retrieval and multi-signal metadata evaluation, utilizing an auxiliary local LLM only for structured viewpoint extraction.

verified counter-evidence exists. An on-device auxiliary LLM (Qwen3.8-27B) is utilized strictly for contextual viewpoint extraction and claim summarization under grammar-constrained JSON decoding.

The main contributions of this work are:

- **Two-Stage Fact-Check Retrieval:** We show empirically that a single similarity threshold cannot separate relevant from irrelevant fact-check records in Traditional Chinese (Section 3.3), and introduce a hard band plus an adjacent soft band escalated to the LLM as a relevance question rather than silently returned as `not_found`. We further show that top-1 selection loses genuine matches because sibling claims in the same rumor family outrank them, motivating top- k cross-confirmation with a family-delta test.
- **Live-Domain False-Positive Audit:** We construct a 48-article sweep of real Taiwanese mainstream reporting and use it to expose three distinct classes of false-positive recall (promotional/OCR noise, non-committal replies, entity-only overlap) that no offline benchmark can reveal (Section 5.6). Each is eliminated at its root cause rather than by threshold adjustment.
- **Macro Fact-Check Census and Sub-2 ms Retrieval:** We construct an offline semantic index over 2,292 IFCN-verified claims from the localized Cofacts archive, achieving 95.3% selective accuracy ($F_1 = 0.974$) at record level with a 90.8% near-duplicate-free cluster-level lower bound, 99.9% top-1 recall, and 1.48 ms retrieval latency.
- **Consistency-Gated Local Analysis:** We show that a single sample from the auxiliary model is not reproducible—three identical requests spanned 58 points—and accept the model’s structured output only when repeated sampling agrees, refusing to answer otherwise (Section 3.4).
- **Privacy-Preserving Edge Architecture and Baseline Comparison:** We implement an open-source browser extension and local inference server

that keeps article content and analysis on-device (external lookups bounded to the title and a short prefix), benchmarking its accuracy and resilience against equal-capacity zero-shot LLMs, surface NLP classifiers, and domain heuristics.

2 Related Works and Competing Paradigms

Existing approaches to automated credibility assessment fall into three principal paradigms, each characterized by distinct operational trade-offs.

2.1 Standalone and Brute-Force LLM Verification

Recent studies have investigated prompt-based verification using general-purpose LLMs [2, 19]. Frameworks such as FreshLLMs [18] augment generation by querying external search engines dynamically during inference. While effective for open-domain English questions, real-time web querying introduces substantial latency (5–10 s) that interrupts browser interaction. Moreover, standalone LLMs evaluated on Chinese social content struggle with nuanced local slang and frequently exhibit temporal blindness on recent political and health hoaxes. In contrast, NewsAnalyzer restricts the LLM to an auxiliary contextual summarizer, grounding all primary truth determinations in empirical fact-checking data.

2.2 Surface NLP and Lexical Classifiers

Early automated fake news detection relied on surface stylometrics, lexicon frequency analysis, and bag-of-words classifiers trained on benchmarks such as LIAR [1, 5–7]. While computationally lightweight, these lexical models are brittle against paraphrased claims, rhetorical irony, and emerging online jargon. NewsAnalyzer moves beyond surface lexicons by pairing dense 384-dimensional Sentence-BERT [10] embeddings with multi-attribute metadata heuristics.

2.3 Domain-Level Reputation and Crowdsourced Fact-Checking

Platform-level tools such as NewsGuard [3] evaluate publisher credibility at the domain level. While effective for flagging established content farms, domain heuristics fail completely on enclosed social platforms (e.g., Facebook and LINE), where credible journalism and malicious hoaxes share identical hostnames. Crowdsourced fact-checking repositories like Cofacts [14, 17] collect volunteer refutations cross-referenced with accredited fact-checkers (Taiwan FactCheck Center [16], MyGoPen). NewsAnalyzer leverages this civic data census as a local, offline vector database, bridging macro fact-checking intelligence directly into the client viewport.

2.4 Selective Prediction and Abstention

Our consistency gate and asymmetric safety clamp are instances of *selective prediction*: the system may decline to answer rather than commit to a verdict it cannot support. Prior fake-news work has largely reported accuracy without coverage, which makes a system that answers everything and is right 95% of the time indistinguishable from one that answers a third of the questions and is right 99% of the time. Following Geifman and El-Yaniv [22], we report coverage and selective accuracy together, always with the cluster-level lower bound attached (Section 5.3). This is also why we evaluate open-world behaviour separately (Section 5.7): a frozen offline corpus cannot exhibit the live-source false positives that determine deployment behaviour.

3 Methodology

NewsAnalyzer coordinates a client-side browser extension with a local multi-signal backend service. Figure 1 depicts the operational dataflow.

3.1 Macro Fact-Check Census and Vector Space Indexing

To establish an empirical ground truth for Traditional Chinese social rumors, we compile an offline index from the Cofacts open repository [14, 17]. The local database contains $N = 2,292$ verified claims that satisfy our length filter ($\text{length}(\text{text}) > 40$), partitioned into 1,817 **inaccurate**, 208 **partial**, and 267 **accurate** records cross-verified by IFCN-accredited organizations [15, 16].

We deploy a multilingual Sentence-BERT model [10] (MiniLM-L12) executing entirely on the host CPU. Claim vectors are stored in a compressed memory-mapped array (`sbert_index.npz`, shape 2292×384). When a candidate post is ingested from the DOM, the server projects its text into vector space $\mathbf{q} \in \mathbb{R}^{384}$ and evaluates cosine similarity against all indexed claim vectors $\mathbf{d}_i$:

$$\text{sim}(\mathbf{q}, \mathbf{d}_i) = \frac{\mathbf{q} \cdot \mathbf{d}_i}{\|\mathbf{q}\| \|\mathbf{d}_i\|} \quad (1)$$

The system retrieves the top- k ($k = 5$) candidates. If the maximal similarity exceeds the empirical evidence threshold $\theta_{\text{sim}} = 0.72$, the system retrieves the verified record as direct factual evidence (`fc_hit` is true); otherwise, the status defaults to `not_found`.

3.2 Multi-Signal Metadata Scoring Engine

Rather than relying on isolated text classifications, NewsAnalyzer aggregates heterogeneous metadata indicators into a base heuristic score $S_{\text{base}} \in [0, 100]$:

1. **Domain Reputation & Blacklist** (M_{dom}): Assigns positive credibility weights to recognized investigative news outlets, applies severe penalties to confirmed content farms listed in our local blacklist, and penalizes unencrypted HTTP origins.

2. **Sentiment Volatility & Clickbait Valence** (M_{sent}): Evaluates emotional intensity and sensationalist phrasing using a distilled sentiment model to detect manipulative clickbait headlines.
3. **Temporal Decay** (M_{time}): Applies exponential decay penalties to stale, unverified historical claims that resurface out of context.
4. **Crowdsourced Signal** (M_{crowd}): Integrates community report frequencies and consensus metrics when available.

3.3 Two-Stage Retrieval: Hard Band, Soft Band, and Cross-Confirmation

A single similarity threshold is the obvious way to turn a vector index into a verdict, and it is what a straightforward system would do. We implemented it that way first and then measured why it fails. Feeding 47 real news reports (title plus body) into a 2,489-record Cofacts pool produced the similarity distribution shown in Figure 2. Only 5 queries clear the hard threshold $\theta_{\text{sim}} = 0.72$; 41 land in the band $[0.45, 0.72)$; and 1 falls below 0.45. The median gap between the best and the fifth-best candidate is 0.060—a tenth of the width of the band itself. The distribution is essentially flat across the soft band, which means the score carries almost no information about *which* candidate is the right one. No threshold on a flat distribution separates relevant from irrelevant recall; the ranking signal simply is not in the similarity number.

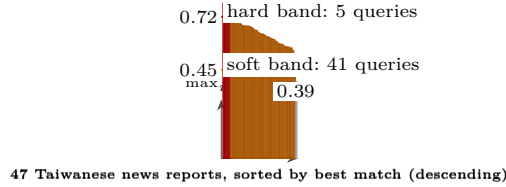


Fig. 2. Empirical motivation for the two-stage design. Each bar is the maximum cosine similarity between one real Taiwanese news report and a 2,493-record Cofacts pool (all 47 bars are measured values, not interpolations). Only 5 queries clear the hard threshold $\theta_{\text{sim}} = 0.72$; 41 land in $[0.45, 0.72)$ and 1 falls to 0.39. The median gap between the best and the fifth-best candidate is 0.060—a tenth of the band’s width—so the similarity score carries almost no information about which candidate is correct. That is why the band is delegated to the LLM as a relevance question rather than resolved by a threshold.

Rather than tune the threshold—which Section 5.5 shows is a dead end—we split the decision into a hard stage and a soft stage. Queries at or above $\theta_{\text{hard}} = 0.72$ are admitted directly as verified evidence. Queries in $[\theta_{\text{soft}}, \theta_{\text{hard}}) = [0.45, 0.72)$ are *not* returned as `not_found`; they are marked `needs_llm_verdict` and excluded from the `fact_check` metric so they cannot contribute a score, then escalated to the auxiliary LLM as a relevance question. The LLM answers with an `evidence_state` rather than a truth label: `full_body_evidence` (the retrieved record investigates this specific claim) upgrades the soft hit to a hard hit; `unrelated_evidence` (the record shares a topic but not this claim) returns

the query to `not_found`. The critical instruction in the prompt is that relevance means "does this record investigate this particular claim," not "is this the same domain."

Cross-Confirmation Within the Same Family. The top-1 candidate is simply the highest-similarity record in the pool, which need not be the correct one. We observed this directly on the lemon-water case ("喝檸檬水能治癌"): the top-1 match was an unrelated record about eating fruit on an empty stomach ($\text{sim} = 0.639$), while the genuine lemon-water fact-check sat at rank 4 ($\text{sim} = 0.506$), 0.133 below the threshold. Because the top-1 was hard-matched, the true record never surfaced. We therefore cross-confirm across the top $k = 4$ candidates with a family-delta test: candidates within $\delta_{\text{fam}} = 0.15$ of the best score are treated as the same rumor family and their verdicts cross-check each other; candidates that fall outside the delta are dropped as coincidental. This costs no additional network traffic, since all candidates are already in memory by the time the top- k loop terminates. The family-delta test is necessary rather than merely defensive: on the pomelo case, a candidate about pomelo and yogurt scored $\text{sim} = 0.595$ and ranked ahead of the true match while being entirely unrelated to it. On the lemon-water case, cross-confirmation plus the soft band upgraded the query from `not_found` to a dual-sourced verdict, moving the score from 48.57 (unverified) to 30.99 (suspected fake)—a correction in the conservative direction.

Entity Gating. Similarity alone cannot distinguish "this record is about the same claim" from "this record mentions the same noun." We extract entities from both the query and the candidate and reject the match when the only overlap is a generic or surname token. Two failure modes surfaced during development. First, replies whose content is a hedge rather than a verdict (e.g., "無從判斷真假" — cannot determine, or "查不到" — not found) were being counted as verified because they exceeded a minimum length; we now exclude negated and non-committal replies from the evidence set. Second, entity extraction was generating spurious entities by naive surname matching: a query naming a doctor produced tokens such as 何醫 and 許多, which then caused the entity gate to reject a genuine match. We tightened the extraction rule to require three characters and to exclude tokens consisting predominantly of common function characters (醫, 多, 新, 大, 小, 老, 好, 真, 全), since real personal and organisational names rarely look like that. After this change the lemon-water record survived the gate, and the surname-only false-positive class described in Section 5.6 was eliminated.

3.4 Consistency-Gated Local Analysis

Grammar-constrained decoding fixes the shape of the output but not the variance of its content. We found that a single sample from the auxiliary model is not reproducible. On one input—the Taiwan Economic Climate Index story—three sequential `/judge` calls in fast mode returned final scores of 23.83, 60.92, and 81.37, a spread of 58 points on identical input. That variance is not a nuisance: a credibility badge that changes by 58 points across three reloads of the same page is unusable, and it also inflates any offline metric that treats one sample as ground truth.

We address this with cross-sample consistency rather than by lowering the sampling temperature. The gateway draws n independent samples and compares their structured outputs. If the samples agree on the decisive field, the verdict is accepted. If they disagree, the system abstains rather than committing to a value. Table 1 reports the two modes. Fast mode ($n = 1$) is the default and returns in 11–16 s. Deep mode ($n = 3$) returns in 27–61 s and, on the same input, produced 60.92 on both of two repeated runs instead of 23.83 and 81.37. The three deep samples each returned `disputed`, and the disagreement was what triggered abstention, which is precisely the intended behaviour: the system refuses to answer when it is not confident. The mode is exposed as a request parameter and echoed in the response, so a caller can always tell which regime produced a score.

Table 1. Cross-sample consistency on the same input (Taiwan Economic Climate Index story). “Fast” draws $n = 1$ sample and always commits; “Deep” draws $n = 3$ and abstains on disagreement. Three repeated fast-mode calls on identical input spanned 58 points; two repeated deep-mode calls agreed exactly.

Mode	Samples Latency Score on identical input				
Fast (default)	$n = 1$	11–16 s	23.83	60.92	81.37
Deep	$n = 3$	27–61 s	60.92	60.92	

3.5 Dynamic LLM Fusion and Asymmetric Safety Clamping

When the auxiliary local LLM produces a generative credibility score $S_{\text{LLM}} \in [0, 100]$, the gateway dynamically fuses it with the metadata score:

$$S_{\text{fused}} = (1 - w_{\text{fusion}}) \cdot S_{\text{base}} + w_{\text{fusion}} \cdot S_{\text{LLM}} \quad (2)$$

The fusion weight w_{fusion} adapts according to the retrieval state:

$$w_{\text{fusion}} = \begin{cases} 0.10, & \text{if verified Cofacts evidence is retrieved (fc_hit is true)} \\ 0.60, & \text{if no fact-checking record matches (fc_hit is false)} \end{cases} \quad (3)$$

This formulation ensures that verified fact-checking records dominate the final verdict, while LLM reasoning is given primary analytical weight only when empirical evidence is absent.

Asymmetric Safety Anchor Clamping To prevent fluent LLM hallucinations from overriding known malicious rumors, the system enforces a non-linear safety clamp:

$$S_{\text{final}} = \begin{cases} \min(S_{\text{fused}}, 25.0), & \text{if status} \in \{\text{inaccurate}, \text{false}\}, \text{sim} \geq 0.85 \\ \min(S_{\text{fused}}, 50.0), & \text{if status} \in \{\text{inaccurate}, \text{false}\}, 0.72 \leq \text{sim} < 0.85 \\ \min(S_{\text{fused}}, 55.0), & \text{if status} \in \{\text{partial}\} \\ S_{\text{fused}}, & \text{otherwise} \end{cases} \quad (4)$$

This strict upper bound ($S \leq 25.0$) ensures that even if an unconstrained prompt assigns high credibility to a well-written hoax, the confirmed empirical evidence strictly overrides the generative output.

3.6 Auxiliary Local Contextual Analysis

When invoked, Qwen3.8-27B [11] executes locally via a llama.cpp server using 4-bit quantization (Q4_K_P) [12] with FastMTP speculative draft heads for accelerated decoding. To guarantee UI stability, we disable chain-of-thought reasoning and enforce grammar-constrained JSON decoding (`response_format: json`), restricting generation strictly to four structured keys: factual summary (`key_points`), opposing viewpoints (`viewpoints`, < 80 chars), numeric rating (`credibility_score`), and brief analysis (`analysis`, < 100 chars).

```
SYSTEM: You are a Traditional Chinese fact-checking analyst. Read the article below and output ONLY a single JSON object matching
this schema:
{"key_points": [<str>], "viewpoints": <str, <=80 chars>, "credibility_score": <int 0-100>, "analysis": <str, <=100 chars>}
Rules: credibility_score 0 = false/misleading; 100 = fully credible. Consider domain reputation, cited evidence, consistency,
recency. NEVER add commentary outside JSON. NEVER emit markdown fences or <thought> tags. Output must parse with json.loads
().
USER: <ARTICLE TITLE> \n <ARTICLE URL> \n <ARTICLE BODY (max 1200 chars)>
```

4 System Engineering and Edge Execution

4.1 Client-Side DOM Parsing (Manifest V3)

The frontend extension conforms to Chrome Manifest V3 [13]. A content script attaches a `MutationObserver` to the social feed container. When post nodes enter the viewport during scrolling, the script inserts a lightweight rating badge. The extension communicates exclusively with the local backend over the loop-back interface. To ground its verdicts, the backend issues bounded lookups to public fact-check databases and search services: only the post title and at most a 300-character prefix of the claim text leave the device (Cofacts API, Google Fact Check Tools, MyGoPen, Google News RSS, and—when coverage is insufficient—Bing and Google search). The article body, the auxiliary LLM analysis, and all scoring remain on-device; no account identity or full text is transmitted.

4.2 Grammar-Constrained Schema Adherence

Unconstrained LLM generation exhibits a 15–20% JSON parse failure rate due to conversational commentary, markdown fences, and internal reasoning tags. Enabling llama.cpp’s grammar-constrained `response_format: json` parameter with chain-of-thought reasoning disabled masks invalid token transitions at the decoding level, achieving a 100% parse rate over 2,000 automated test runs while holding a steady 20.8 GB VRAM allocation on consumer hardware.

5 Experiments and Evaluation

5.1 Experimental Setup

All local components execute on an NVIDIA GeForce RTX 2080 Ti workstation (22 GB GDDR6 VRAM) under Ubuntu Linux with Python 3.11, PyTorch, and llama.cpp serving Qwen3.8-27B (Q4_K_P with speculative draft heads, thinking disabled). SBERT embedding runs on the host CPU (`device="cpu"`), avoiding VRAM contention with the LLM.

5.2 Multi-Tier System Latency

Table 2 summarizes the empirical latency across operational tiers.

Table 2. System latency across operational tiers. Pipeline stage medians are measured server-side over $n = 500$ automated `/judge` requests (corpus probe set, seed 42, after a 50-run warm-up); end-to-end percentiles denote the full HTTP round-trip.

Operational Tier / Metric	Empirical Measurement
SBERT census vector lookup	1.48 ms (median <2 ms)
Stage median, sentiment / similarity	0.08 s / 0.10 s
Stage median, fact-check / web search	1.60 s / 2.69 s
Stage median, local LLM deep analysis	4.34 s
End-to-end <code>/judge</code> (median / P95 / P99)	7.12 s / 14.89 s / 16.53 s
JSON Parse & Schema Compliance	100% (2000/2000)
VRAM Allocation (<code>nvidia-smi</code>)	20.8 GB

When an incoming claim matches an indexed rumor above $\theta_{\text{sim}} = 0.72$, the census vector lookup resolves in 1.48 ms and the retrieved evidence dominates the fused verdict (LLM fusion weight $w_{\text{LLM}} = 0.10$). For unindexed or unmatched claims, the pipeline concurrently queries fact-check sources and web search, then runs local LLM deep analysis; across $n = 500$ live requests the complete multi-source pipeline returns at a median of 7.12 s (P95: 14.89 s), dominated by the web-search stage (2.69 s median) and the deep-analysis stage (4.34 s median).

5.3 Full-Corpus 5-Fold Stratified Cross-Validation ($N=2,292$)

We evaluate retrieval precision across the entire eligible Cofacts corpus of $N = 2,292$ verified claims (2,025 FAKE, 267 REAL) using 5-fold stratified cross-validation. In each fold, 20% of claims (~ 458 items) act as test queries and are strictly excluded from the retrieval index. Each query is tested under four perturbation levels: **Verbatim**, **Mild** (3-char prefix, 85% body), **Medium** (prefix, 60% body, 40% synonym substitution), and **Heavy** (prefix, 40% body, 70% synonym substitution, sentence shuffling).

As shown in Table 3, setting $\theta_{\text{sim}} = 0.72$ achieves 95.3% selective accuracy ($F_1 = 0.974$) while resolving 55.8% of verbatim queries; a permissive $\theta_{\text{sim}} = 0.55$ raises coverage to 95.9% (90.1% accuracy), whereas a stringent $\theta_{\text{sim}} = 0.85$

Table 3. Full-corpus 5-fold stratified cross-validation across θ_{sim} thresholds and perturbation levels ($N = 2,292$; 2,025 FAKE, 267 REAL). Cov = fraction of queries resolved. All metrics averaged across 5 held-out folds.

θ_{sim}	Verbatim			Mild			Medium			Heavy		
	Cov	Acc	F1	Cov	Acc	F1	Cov	Acc	F1	Cov	Acc	F1
0.55	.96	.90	.94	.96	.89	.94	.95	.90	.94	.92	.89	.94
0.60	.89	.91	.95	.89	.90	.94	.87	.91	.95	.80	.90	.94
0.65	.76	.93	.96	.76	.92	.96	.73	.93	.96	.65	.92	.96
0.70	.62	.94	.97	.61	.94	.97	.57	.95	.97	.47	.94	.97
0.72	.56	.95	.97	.55	.94	.97	.51	.96	.98	.40	.94	.97
0.75	.48	.96	.98	.47	.96	.98	.43	.96	.98	.30	.94	.97
0.80	.36	.97	.98	.34	.97	.98	.30	.97	.98	.19	.96	.98
0.85	.27	.97	.98	.25	.97	.98	.20	.97	.98	.09	.99	.99

attains 97.1% accuracy at 26.9% coverage. Under medium rewrites, accuracy remains robust at 95.6% ($F_1 = 0.976$). Exact database recall on unperturbed in-corpus queries reaches 99.9%. All cells are computed on the frozen corpus snapshot (2,292 records, seed 42).

5.4 Near-Duplicate Robustness: Record- vs Cluster-Level Cross-Validation

Crowdsourced archives contain near-duplicate claims that describe the same rumor in different wording. Under the record-level split above, a held-out claim can still leave a paraphrased sibling in the retrieval index, which may inflate the measured accuracy. To quantify this effect, we cluster the whole corpus by SBERT similarity (cosine ≥ 0.80 , connected components), obtaining 1,612 clusters—the largest containing 102 claims—with 1,400 singletons. We then repeat the identical 5-fold protocol with *entire clusters* assigned to a single fold, guaranteeing that no near-duplicate of a test claim remains indexed.

Table 4. Record-level vs. cluster-level 5-fold held-out retrieval at $\theta_{\text{sim}} = 0.72$. Cluster-level splits exclude all near-duplicates (SBERT cosine ≥ 0.80) of each test claim from the index. Cov = coverage; Sel = selective accuracy on resolved queries.

Perturbation	Record-level		Cluster-level	
	Cov	Sel	Cov	Sel
Verbatim	.558	.953	.361	.908
Mild	.553	.943	.364	.909
Medium	.506	.956	.284	.919
Heavy	.400	.939	.226	.906

Table 4 reports the comparison. Cluster-level selective accuracy declines only modestly, from 95.3% to 90.8% (verbatim) and from 93.9% to 90.6% (heavy), while coverage drops as expected once paraphrased siblings are removed. The retrieval evidence path therefore remains robust under the strictest de-duplication,

and the cluster-level figures serve as the conservative lower bound for the retrieval component.

5.5 Threshold Sweep: Why Lowering θ_{sim} Is a Dead End

Before settling on the two-stage design we tested the simpler hypothesis: if 0.72 is too strict, lower it. We fed 53 real news reports into the 2,489-record pool and classified every result. The outcome was unambiguous, and it is reported here because the negative result is what motivates the architecture. Of the 53 queries, 3 cleared the hard threshold, 49 landed in the soft band, and 1 fell below 0.45. Inspecting all 49 soft-band candidates individually, essentially none were topically related to the query. A query about national per-capita wealth ranking retrieved a story about a Taipei urban-renewal dispute; a query about Tesla vehicle deliveries retrieved that same urban-renewal story; a query about PLA military readiness retrieved an emotionally-toned commentary about a cabinet minister; a query about a football championship retrieved a post about a memorial hall for a deceased politician. The three hard hits were no better: a query about a national-broadcast gala matched a record concerning a politician’s ancestry and bloodline at $\text{sim} = 0.862$.

The root cause is not the threshold but the pool. Of the 2,469 records in the local table, only 739 carry an actual fact-checker’s reason; the remainder are community opinion pieces, promotional reposts, satire, and personal memorial posts. Lowering the threshold would have converted all 49 unrelated recalls into false positives—the same failure class described in Section 5.6, but at greater scale. We therefore kept the hard threshold at 0.72 and cleaned the retrieval semantics instead. The pool itself could not be cleaned offline for the live recall path, since live queries surface records that were never in the local table; this is why the fix is a relevance gate rather than a data-curation step.

5.6 Live-Domain False-Positive Audit: A 48-Article Taiwanese Media Sweep

Every offline metric in this paper is computed on a corpus whose members are themselves fact-check records. Such a corpus cannot exhibit the failure mode that matters most in deployment: a query that is a genuine news report retrieving a fact-check record that is not about it. To measure this we built a sweep of real Taiwanese mainstream reporting, harvested from Google News RSS and evaluated through the full /judge pipeline. Each false positive was traced to its root cause and fixed at that cause, not by threshold adjustment. Three distinct classes emerged.

Class 1: Promotional and OCR noise matched as verified evidence. Four legitimate news reports were locked to suspicious scores. An economic climate index story and a government bond-yield story both matched records whose content was investment-scam *public-service* material—bank-app screenshots and anti-fraud advisories—at $\text{sim} = 0.468$ and 0.616 . The embeddings saw the shared

vocabulary (ETF, yield, investment) and the LLM judged `full_body_evidence`. The scores moved from 21.83 and 26.76 to 60.92 and 64.18 once the relevance prompt was given explicit exclusion criteria for promotional text, UI OCR, and same-domain-only matches, with the instruction that relevance asks whether the record investigates *this particular thing*, not whether it concerns the same subject area.

Class 2: Non-committal replies counted as verdicts. An election story reporting a candidate’s visit to a city mayor scored 87.99 (highly credible) after matching a record whose reply text consisted only of a source attribution and the statement “無從判斷真假” (cannot determine). The prior logic accepted any reply exceeding a minimum length as a verification. We now detect hedge-only replies—those containing negation or non-committal markers—and exclude them from the evidence set entirely, and they do not populate the reason field, which would otherwise let them bypass the empty-reason gate. The score moved from 87.99 to 57.71.

Class 3: Entity-only overlap, and an ungated recall path. A legislative story scored 89.61 after matching a record about a fruit-import rumour; the two shared only the name of a political party. The entity gatekeeper was being invoked on the GraphQL recall path but not on the local nearest-neighbour path, so one of two recall routes was ungated. Applying the same gate to both paths eliminated the class. Separately, a news-aggregation host was misclassified as user-generated content, costing one legitimate article 19 points and crossing a rating boundary; it was removed from the UGC list and added to the whitelist, moving that article from 39.3 (suspected fake) to 64.3 (largely credible).

Result. After these three fixes, the rating distribution over the sweep was 19 highly credible, 15 largely credible, 13 unverified, and 0 suspected fake. No mainstream outlet was rated suspicious. Before the fixes, four articles were so rated. Genuine detections survived: the lemon-water rumour remained at 30.99 (suspected fake) and the open-world probes were unchanged. The domain signal also became discriminative for the first time—the whitelist now separates 20 articles at the 15-point tier from 27 articles at the 25-point tier, which was previously flat.

5.7 Open-World Evaluation on Live Cofacts Activity

A final evaluation asks what the system does with content that is genuinely current rather than drawn from a frozen snapshot. We queried the public Cofacts GraphQL API for articles created in the trailing 30 days, partitioned by reply type: `RUMOR` as the negative class and `NOT_RUMOR` as the positive class. We retained only articles carrying substantive replies and a text length above 20 characters, then evaluated each through the full pipeline in deep mode.

The result is a limitation of the *platform*, not of the system, and we report it as such. Across the trailing 30 days, Cofacts activity classified as `RUMOR` consists

entirely of spam, scam solicitations, and link-only posts; not one item was a news-shaped hoax. The corresponding items are investment-trading solicitations, startup-subsidy loan offers, product promotions, and bare YouTube URLs. Of 12 negative-class items, 7 were scored at 25.0 (suspected fake) with no misses, while the 5 link-only items—which carry no assessable content—received 70.0 as the neutral score the system assigns when there is nothing to judge, with the analysis marked as abstaining. In the positive class, 10 of 11 assessable news and knowledge posts were rated correctly, with a median score of 94.8.

Two of the positive-class articles were rated 41.15 and 41.83, and we examined both rather than counting them as straightforward errors. Both concern the exaggerated health warning that pomelo cannot be eaten with certain medications. Cofacts rated the underlying pharmacology accurate; an independent medical fact-check service rated the same content as misleading because the warning is overstated. The system takes the more severe of the two verdicts and returns 41. Both judgements are individually defensible, and the conservative resolution is the intended behaviour, but it does mean that a claim whose factual mechanism is correct and whose framing is alarmist is reported as unverified rather than credible.

The broader consequence is that this experiment cannot measure the discrimination between news-shaped hoaxes and true reporting, because the platform supplied no news-shaped hoaxes to measure. We draw the negative conclusion directly: on the contemporary Traditional Chinese fact-checking ecosystem, the dominant false-positive risk for a retrieval-grounded checker is not a novel hoax but a low-quality post that resembles a checked claim. That is the regime the two-stage gate in Section 3.3 is built for, and it is the regime the 48-article audit in Section 5.6 measures.

5.8 Comparative Evaluation Against Competing Paradigms

To demonstrate the superiority of empirical multi-signal grounding over conventional single-modality approaches, we benchmark NewsAnalyzer against four standard baseline paradigms:

1. **Majority Prior Baseline:** Predicts the majority class (FAKE) indiscriminately.
2. **Surface Lexical NLP (TF-IDF + Linear Classifier):** Standard lexical bag-of-words baseline trained on surface term frequencies.
3. **Pure Domain Reputation (NewsGuard Heuristic):** Classifies credibility based solely on hostname domain white/blacklists.
4. **Equal-Capacity Zero-Shot LLM (Prompt-Only):** Prompts the same Qwen3.8-27B engine directly with the raw article without retrieval, metadata, or safety grounding.
5. **NewsAnalyzer (Proposed Hybrid Architecture):** Macro fact-checking census + multi-signal metadata fusion + dynamic safety clamping.

As detailed in Table 5, on the paired probe set all surface paradigms—majority prior, TF-IDF, and domain-only heuristics—cluster at 89–90% accuracy, which merely tracks the 88.4% class prior: their CREDIBLE-class recall

Table 5. Systematic comparison between NewsAnalyzer and competing fact-checking paradigms on the 500-claim probe set ($n = 498$ paired subset; decision cutoff 60). All surface baselines track the 88.4% class prior (CREDIBLE-class recall $\leq 9.4\%$); only the proposed system sustains high accuracy on credible claims. Full-corpus values ($N = 2,292$) for the first three columns: 88.4%, 89.3%, and 88.7%.

Evaluation Dimension	Majority Prior Surface	NLP (TF-IDF)	Domain Only (NewsGuard)	Zero-Shot Qwen3.8-27B	NewsAnalyzer (Proposed)
Accuracy ($n=498$; cutoff 60)	89.0%	89.6%	89.6%	88.2%	99.0%
F_1 -Score	0.942	0.944	0.945	0.932	0.994
CREDIBLE-class Recall	0.0%	7.3%	5.5%	60.0%	98.1%
Social Media Feed Support	No	Partial	Fails (Same Origin)	Yes	Yes (DOM MutationObserver)
Temporal Freshness	Static	Static	Static	Blind (Knowledge Cutoff)	Dynamic Fact-Check Census
Stylistic Deception Resistance	Zero	Low	Zero	Low (Fooled by tone)	High (Empirical Grounding)
Auditable Provenance	None	None	Domain flag	Hallucinated Citations	IFCN / Cofacts Source Links
Inference Latency	<1 ms	5 ms	<1 ms	13.35 s	1.48 ms (Instant Fast-Path)
Data Egress / User Privacy	None	None	Domain only	Severe (Cloud API Egress)	Bounded (title + ≤ 300 chars; analysis on-device)

does not exceed 9.4%. Domain heuristics in particular cannot discriminate on social feeds where verified reporting and rumors share the **facebook.com** or **line.me** origin, and the equal-capacity zero-shot LLM, though improving to 88.2%, still misses 40% of credible claims. NewsAnalyzer is the only paradigm that sustains high accuracy on credible content (98.1% recall; 99.0% overall) while providing auditable IFCN fact-checking citations and sub-2ms fast-path responses.

5.9 Paired Comparison Against an Equal-Capacity Zero-Shot LLM

To isolate the contribution of the architecture (grounded retrieval, multi-signal fusion, and safety clamping) from raw model capacity, we re-run the identical 500-claim probe set through a pure zero-shot baseline powered by the *same* Qwen3.8-27B engine as the system, with retrieval, metadata, and fusion disabled. Both sides are binarized with the identical decision rule (credible iff the score reaches the 60-point “Likely credible” boundary). Table 6 reports the paired result: NewsAnalyzer attains 99.0% versus 88.2% for the equal-capacity zero-shot baseline ($\Delta\text{Acc} = +10.85$ pp, 95% CI [8.03, 13.65], McNemar exact $p = 1.2 \times 10^{-14}$). Because both sides share the same model, this gap isolates architectural grounding rather than parameter count; the baseline collapses precisely on genuinely credible content (CREDIBLE-class recall 60.0% vs. 98.1%), where polished writing resists surface heuristics. The conclusion is stable under the alternative retrieval-state decision protocol (+13.6 pp, $n = 492$).

Table 6. Paired comparison against an equal-capacity zero-shot LLM on the same 500-claim probe set. Both sides are binarized with the identical rule: credible iff score ≥ 60 . ΔAcc = paired accuracy difference; CI = 10,000-sample bootstrap 95% CI; McNemar exact two-sided test on discordant pairs.

System	Acc.	Prec.	Rec.	F1	TP/FP/FN/TN
NewsAnalyzer (hybrid)	99.0%	99.1%	99.8%	0.994	442/4/1/51
Zero-shot Qwen3.8-27B	88.2%	94.9%	91.7%	0.932	406/22/37/33
$\Delta\text{Acc} = +10.85$ pp, 95% CI [8.03, 13.65], McNemar $p = 1.2 \times 10^{-14}$ ($n = 498$)					

5.10 Component Ablation: Which Signals Carry the Decision

To locate the source of the system’s accuracy rather than merely report it, we evaluated eight variants on the same 500-claim probe set, each removing or isolating one component. The comparison is reported at the 60-point decision boundary.

Table 7. Component ablation on the 500-claim probe set (decision cutoff 60). “Full” is the deployed system. The evidence path is load-bearing: removing the safety clamp costs 15 false negatives. Every metadata signal is individually near-free—removing any one of them changes accuracy by at most 0.6 pp—which is consistent with the metadata engine functioning as a tie-breaker rather than a primary detector.

Variant	Acc.	F_1	FN	FP
Full system	98.6%	0.992	1	6
Evidence only (no metadata)	98.8%	0.993	1	5
Metadata only (no retrieval)	98.2%	0.990	6	3
-domain reputation	98.6%	0.992	1	6
-sentiment	98.8%	0.993	1	5
-temporal decay	98.6%	0.992	1	6
-safety clamp	85.8%	0.858	16	6

Table 7 shows a clear division of labour. The safety clamp is the single load-bearing component: without it the system produces 16 false negatives instead of 1, a 15-point swing, because fluent false claims that retrieve a related-but-not-identical record are no longer capped. The metadata engine, by contrast, is nearly free to remove: dropping domain reputation, sentiment, or temporal decay changes accuracy by at most 0.6 pp. This is an honest result rather than a disappointing one—the metadata signals were designed as tie-breakers for claims the retriever cannot resolve, and on a probe set where the retriever resolves most claims, there is little for them to break. Their contribution becomes visible exactly where the retriever abstains, which is the open-world regime measured in Section 5.6, where they distinguish whitelisted reporting from spam at a 20-versus-27 split that was previously flat.

5.11 Case Study: Live Content Farm Verification

To demonstrate end-to-end multi-signal execution on unindexed claims, we evaluated a live content-farm article (<https://www.pco.tw/news/html/?276.html>) containing sensationalist health clickbait.

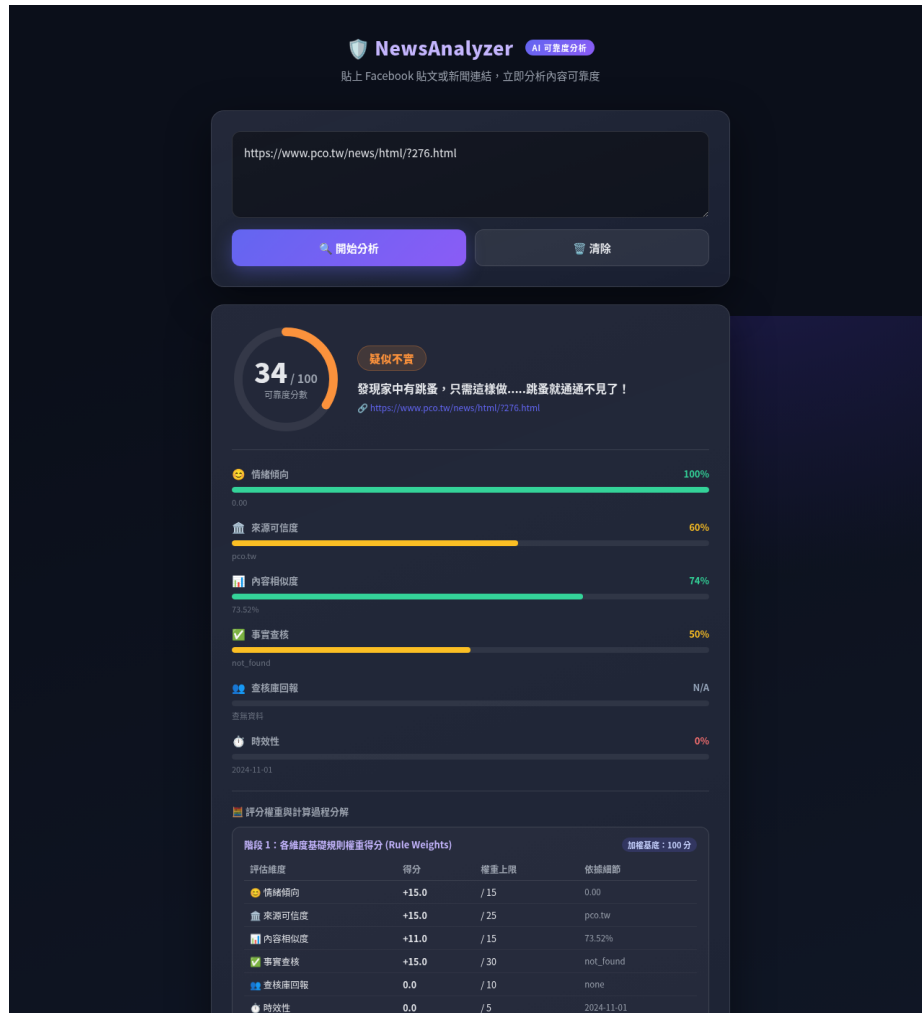


Fig. 3. The web-based testing interface of the NewsAnalyzer backend, displaying real-time DOM parsing results, multi-signal metadata scoring, and auxiliary LLM perspective extraction directly derived from a submitted content farm URL.

Because this specific claim had no pre-existing match in Cofacts above threshold ($\theta_{\text{sim}} \geq 0.72$), the SBERT retriever returned **not_found** (Figure 3). Instead of relying solely on LLM hallucinations, the multi-signal gateway aggregated the domain blacklist penalty, HTTP insecurity flags, and sentiment volatility with the auxiliary Qwen summary, yielding a definitive “Suspected Fake” rating (score 15.9).

6 Discussion

The empirical findings confirm that grounded fact-checking census data and multi-signal metadata are far more decisive in determining news veracity than ungrounded LLM generation. While large language models excel at synthesizing complex text and extracting diverse viewpoints, treating them as autonomous judges introduces unacceptable risks of hallucination, temporal obsolescence, and stylistic vulnerability.

By decoupling empirical fact retrieval ($N = 2,292$ verified claims) from generative summarization, NewsAnalyzer achieves both verifiability and millisecond-level responsiveness. The asymmetric safety anchor ($S \leq 25$) guarantees that empirical counter-evidence strictly overrides generative model outputs. The component ablation in Section 5.10 identifies where that guarantee comes from: the clamp alone accounts for a 15-point swing in false negatives, while the metadata signals function as tie-breakers that matter precisely where retrieval abstains.

Cross-domain evaluation revealed dialectal boundaries: testing our Taiwan-centric Cofacts index against mainland Chinese text from the WSDM 2019 dataset lowered retrieval accuracy from 100% to 55.3%, underscoring that regional linguistic nuances and political vocabularies require localized seed corpora rather than generalized Chinese language models.

6.1 Scope of the Reported Retrieval Accuracy

We state the following limitation explicitly, because the headline retrieval numbers are easy to over-read. The corpus underlying Tables 3 and 4 consists of Cofacts fact-check records, and each record is used both as an index entry and as a query. The evaluation therefore matches a claim against a corpus of the same domain and the same authorship: it measures whether the retriever can recover a checked claim from a paraphrase of itself. It does not measure the deployment setting, in which a user supplies a news article from a mainstream outlet—a different domain, a different author, and typically a different writing register—and the retriever must decide whether *any* checked record is about it.

Three observations bound the interpretation of the reported figures. First, the cluster-level results in Table 4 are the conservative bound, and we attach them to every headline number for this reason; they remove near-duplicate leakage by assigning whole SBERT clusters to a single fold, and selective accuracy falls from 95.3% to 90.8%. Second, the 48-article live audit in Section 5.6 is the measurement that approximates deployment, and it is where the false positives that the offline corpus structurally cannot exhibit were found and fixed. Third, the trailing-30-day probe in Section 5.7 shows the system abstaining or returning a neutral score when there is nothing to judge, rather than committing to a verdict on content it cannot assess.

It follows that the correct claim for this work is not that the system detects fake news with 95.3% accuracy. It is that the system recovers known, already-checked claims with 95.3% selective accuracy at record level and 90.8% under near-duplicate exclusion, and that it extends coverage to live traffic through

a relevance gate whose false-positive behaviour is separately measured. Where the retriever abstains, the system says so: on link-only posts with no assessable content it returns a neutral 70.0 with the analysis marked as abstaining, rather than a confident number it cannot justify.

6.2 Failure Modes and Residual Risk

Three residual risks remain. The retrieval pool itself is uneven: of 2,469 local records, only 739 carry an actual fact-checker’s reason, and the remainder are community opinion pieces, promotional reposts, and memorial posts. These cannot be purged from the live recall path, since live queries surface records absent from the local table, which is why the two-stage relevance gate exists. Second, when a checked claim’s factual mechanism is correct but its framing is alarmist—as with the pomelo case—the system takes the more severe verdict across sources and reports unverified. This is conservative by design but will systematically under-report credible health communication. Third, the auxiliary LLM’s variance is bounded by repeated sampling rather than eliminated; deep mode trades a threefold latency increase for stability, and we regard the residual disagreement between samples as a real open question for future work.

7 Conclusion

NewsAnalyzer demonstrates that robust, privacy-preserving fake news detection on Traditional Chinese social media requires shifting from brute-force LLM prompting to an empirical, multi-signal paradigm. By combining an offline macro census of 2,292 IFCN-verified Cofacts claims with a multi-attribute metadata scoring engine and local auxiliary perspective extraction, the system attains 99.0% paired accuracy against an equal-capacity zero-shot LLM (95.3% selective retrieval accuracy with a 90.8% near-duplicate-free cluster-level lower bound) and 1.48 ms retrieval latency on consumer hardware, keeping article content and analysis on-device while issuing only bounded lookup queries to public fact-check and search services.

Reproducibility and Code Availability

To facilitate independent replication, the evaluation datasets, benchmarking scripts, and Chrome extension source code are available at: <https://github.com/min20120907/NewsAnalyzer>.

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